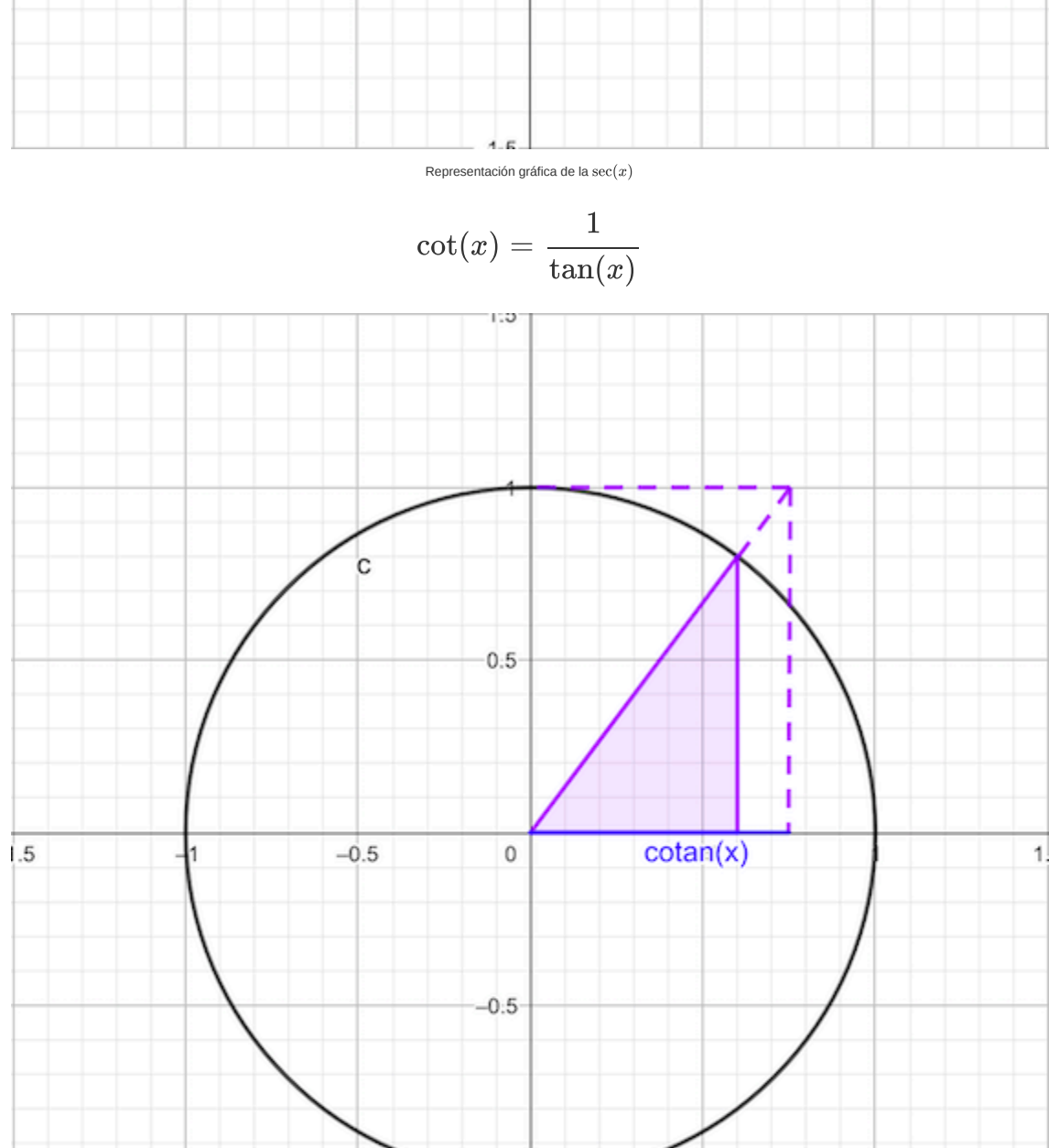
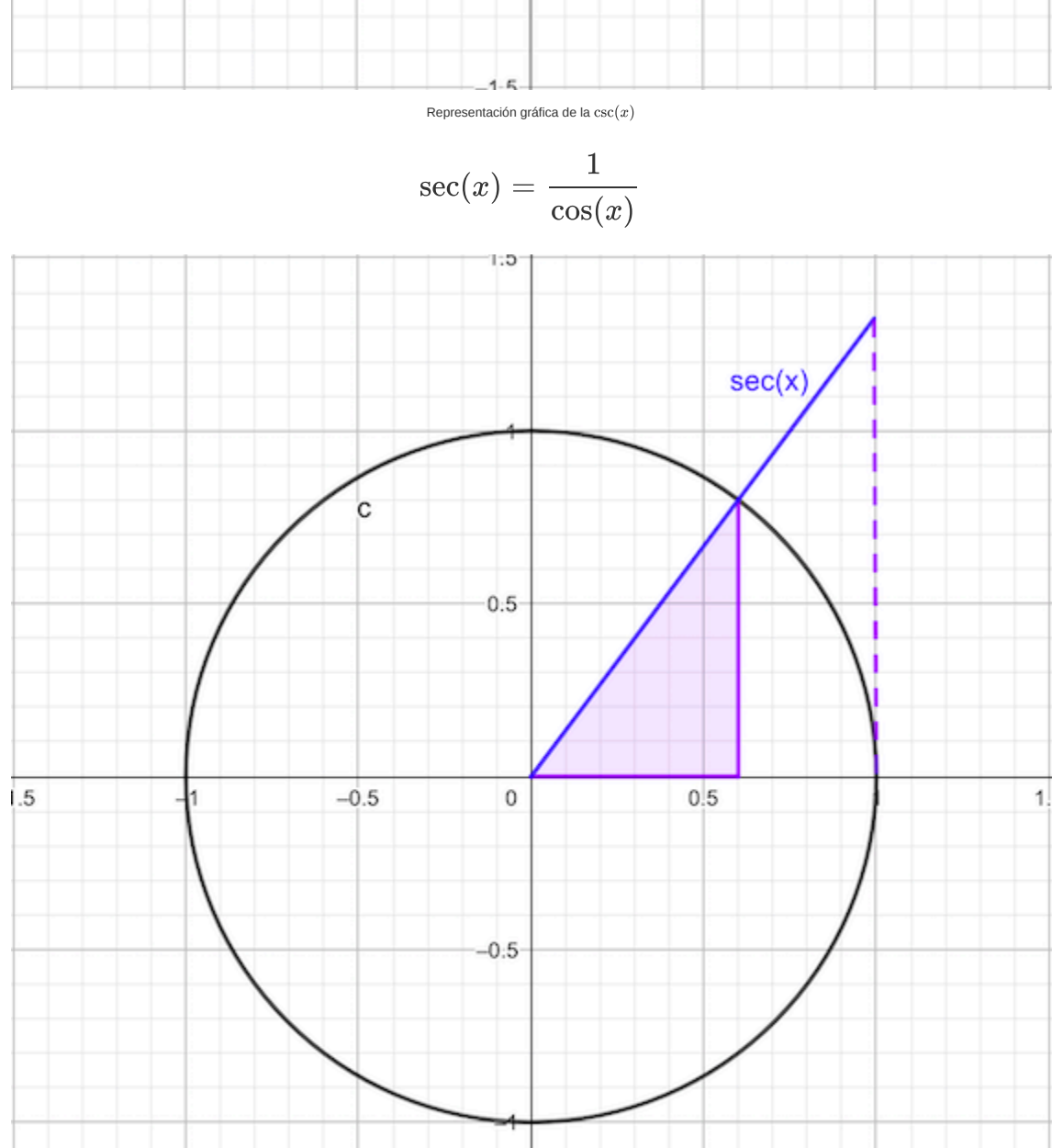
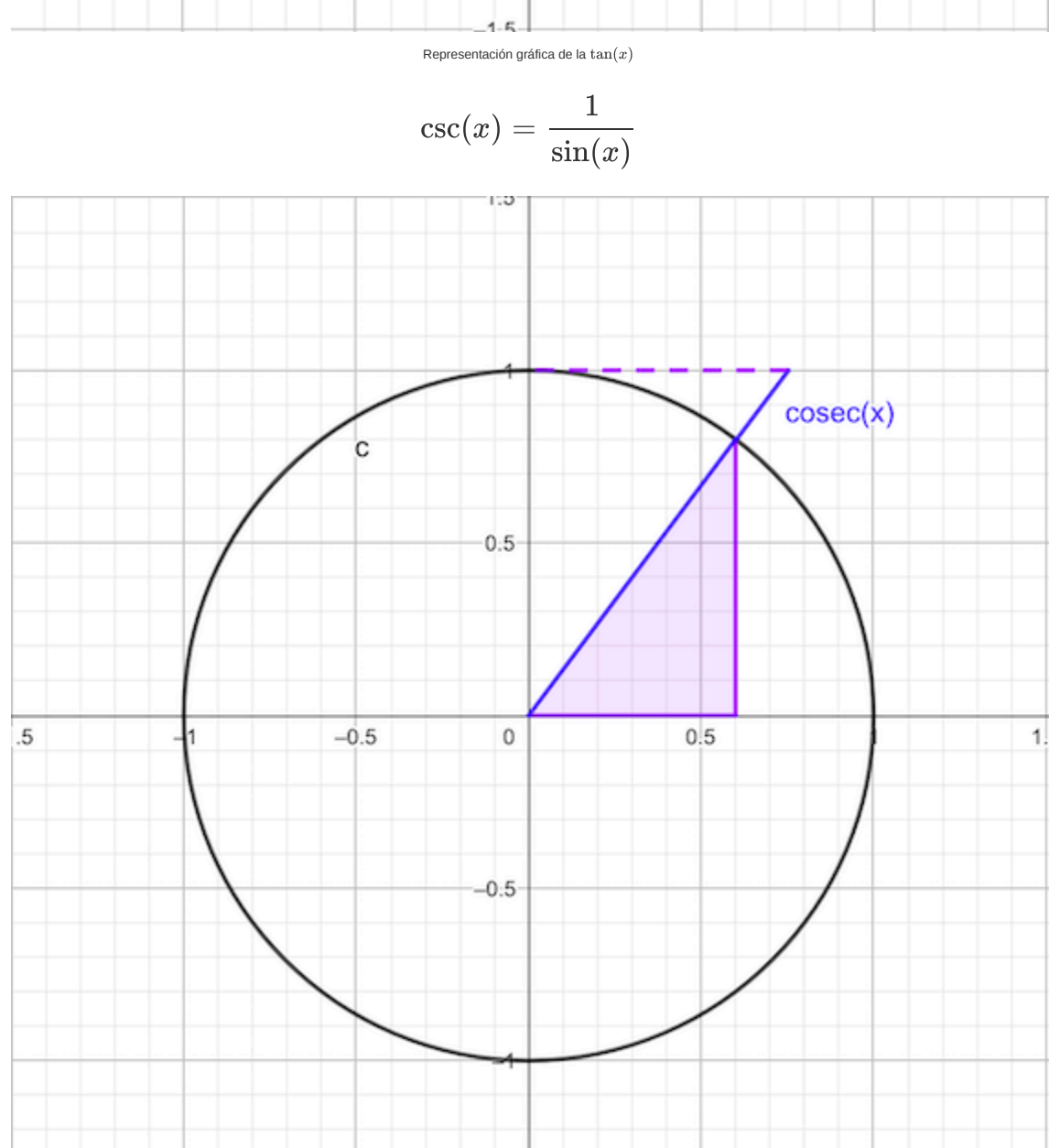
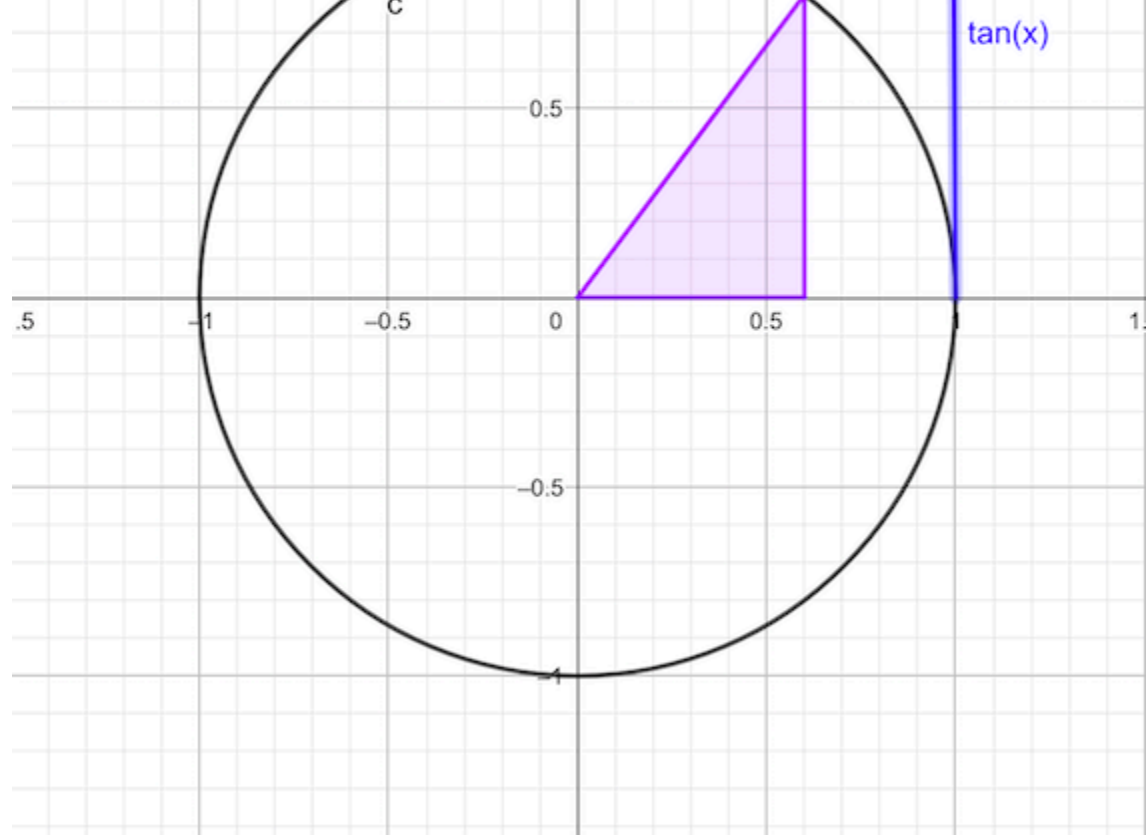
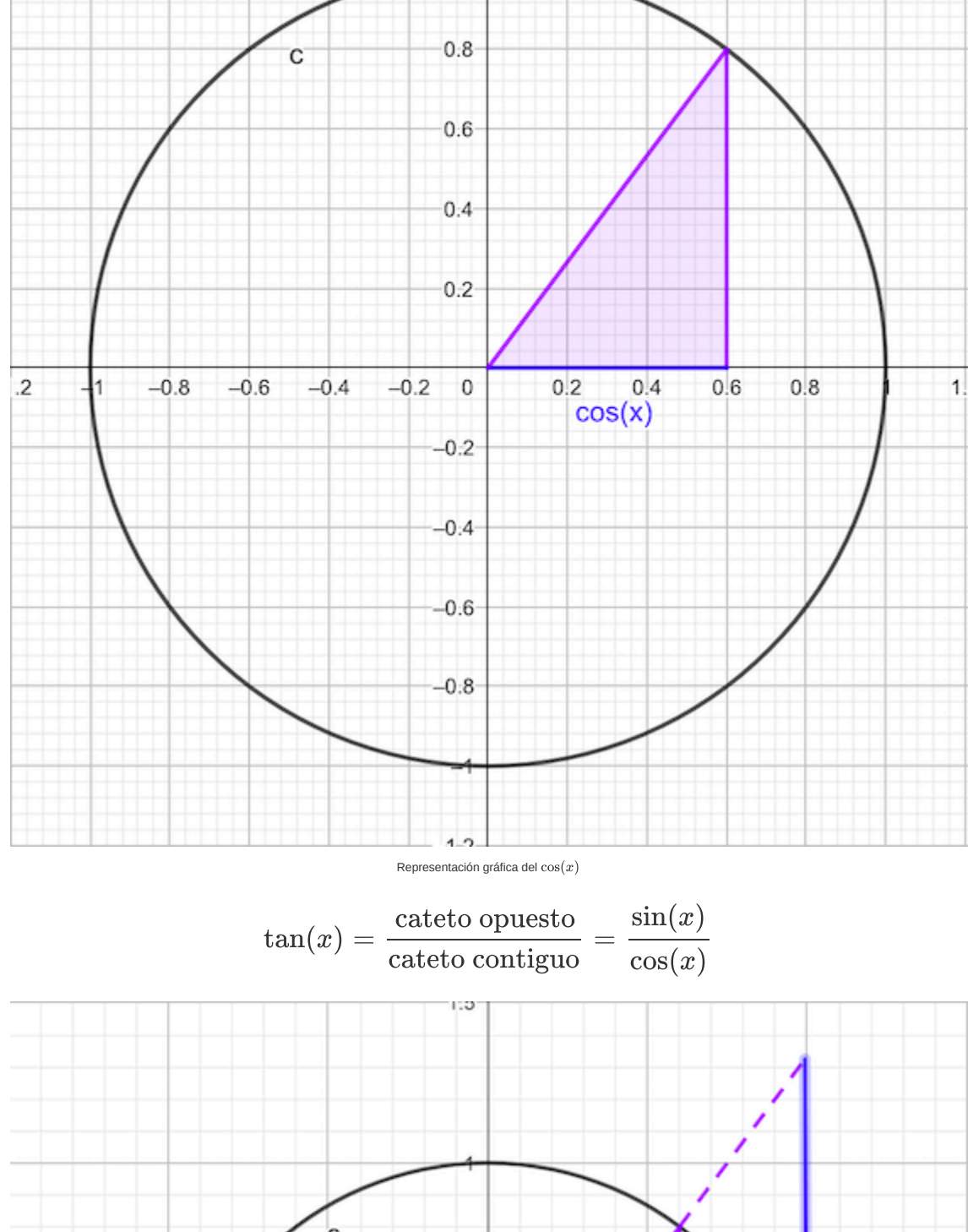
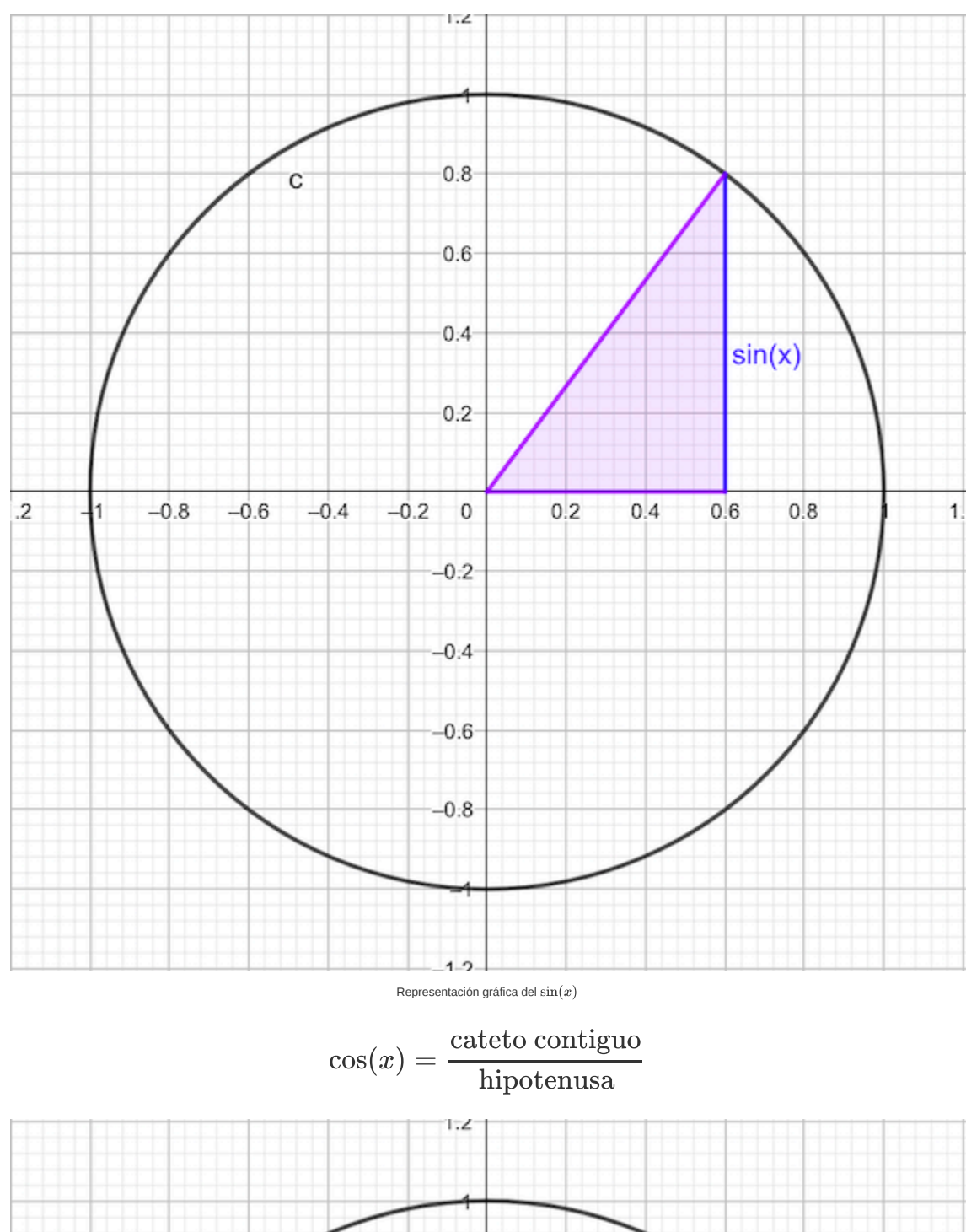


Trigonometría - Resumen de Fórmulas

Curso Álgebra Lineal

Razones trigonométricas



Relaciones fundamentales

$$\sin^2(\alpha) + \cos^2(\alpha) = 1$$

$$1 + \tan^2(\alpha) = \frac{1}{\cos^2(\alpha)}$$

Relaciones pitagóricas

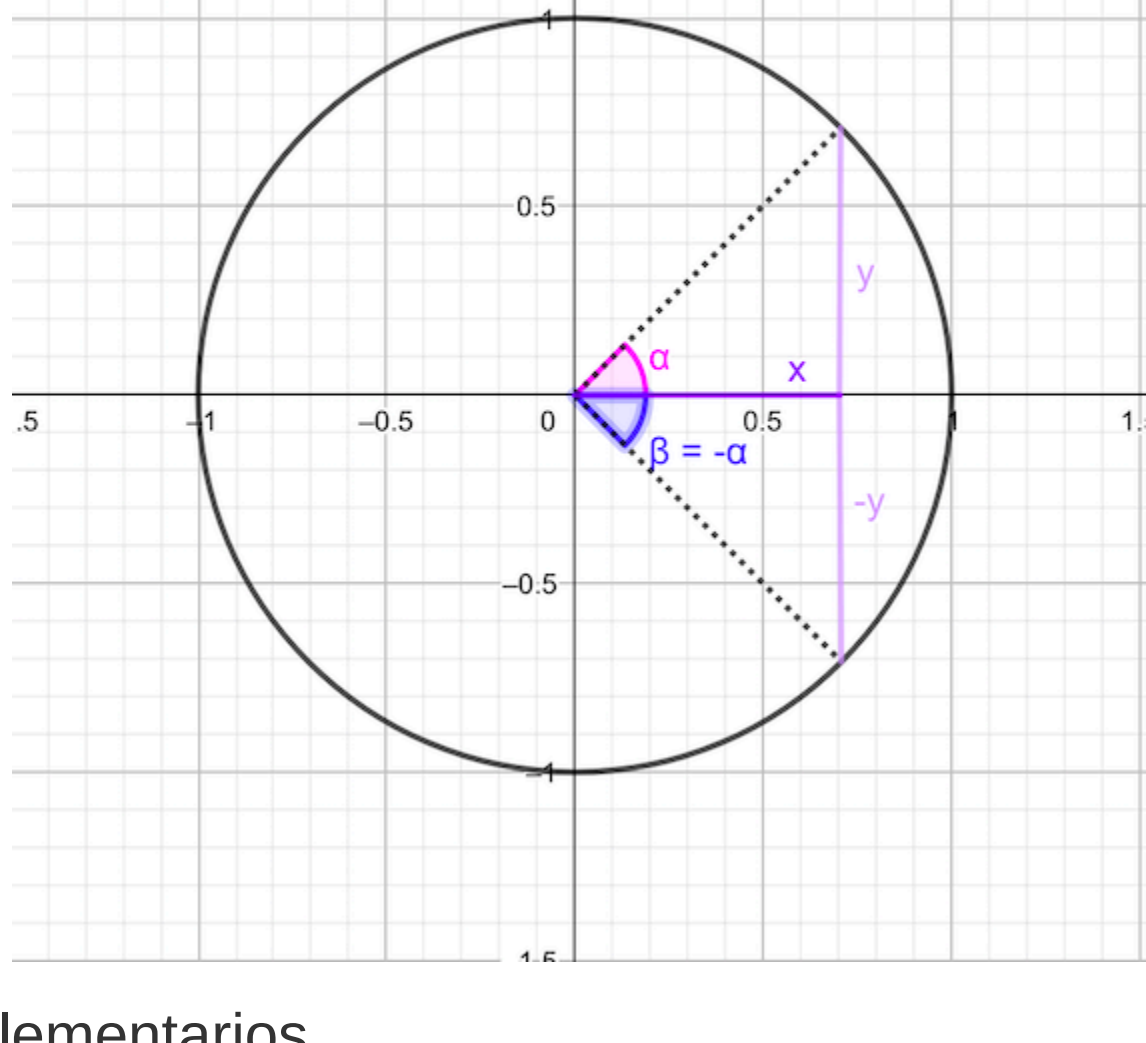
$$1 + \cot^2(\alpha) = \csc^2(\alpha)$$

$$1 + \tan^2(\alpha) = \sec^2(\alpha)$$

Relaciones entre las razones trigonométricas

Ángulos opuestos

$$\begin{aligned} \sin(-\alpha) &= -\sin(\alpha) & \cos(-\alpha) &= \cos(\alpha) & \tan(-\alpha) &= -\tan(\alpha) \\ \sin(2\pi - \alpha) &= -\sin(\alpha) & \cos(2\pi - \alpha) &= \cos(\alpha) & \tan(2\pi - \alpha) &= -\tan(\alpha) \end{aligned}$$

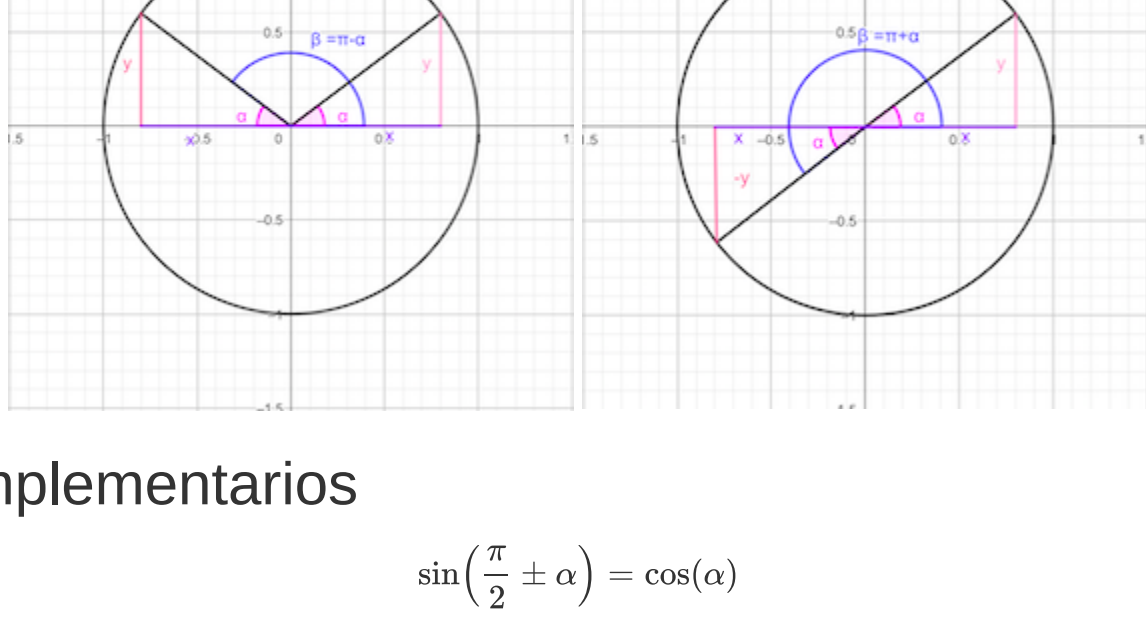


Ángulos suplementarios

$$\sin(\pi \pm \alpha) = \pm \sin(\alpha)$$

$$\cos(\pi \pm \alpha) = -\cos(\alpha)$$

$$\tan(\pi \pm \alpha) = \mp \tan(\alpha)$$

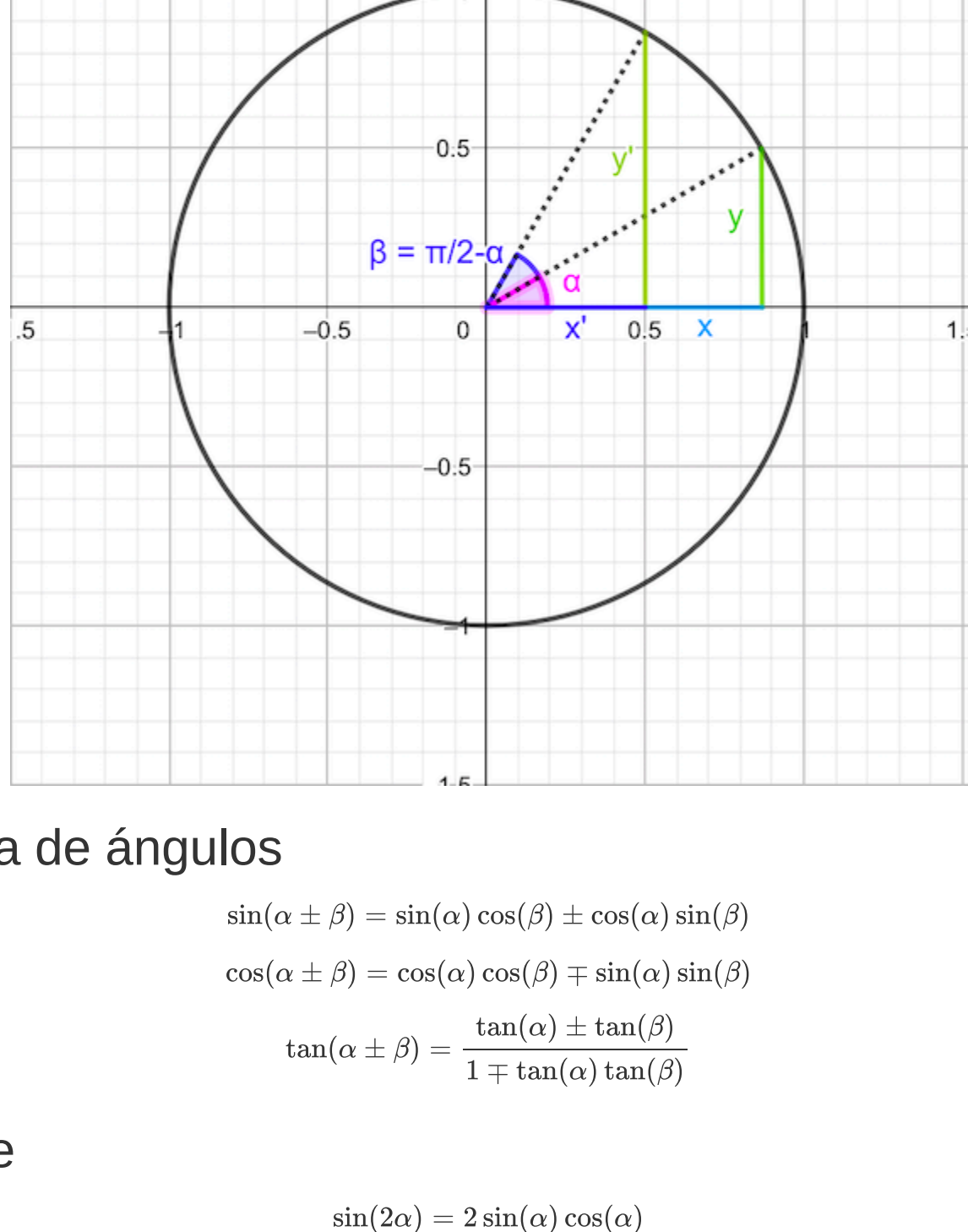


Ángulos complementarios

$$\sin\left(\frac{\pi}{2} \pm \alpha\right) = \cos(\alpha)$$

$$\cos\left(\frac{\pi}{2} \pm \alpha\right) = \mp \sin(\alpha)$$

$$\tan\left(\frac{\pi}{2} \pm \alpha\right) = \mp \frac{1}{\tan(\alpha)}$$



Suma o resta de ángulos

$$\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\alpha) \sin(\beta)$$

$$\tan(\alpha \pm \beta) = \frac{\tan(\alpha) \pm \tan(\beta)}{1 \mp \tan(\alpha) \tan(\beta)}$$

Ángulo doble

$$\sin(2\alpha) = 2 \sin(\alpha) \cos(\alpha)$$

$$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha)$$

$$\tan(2\alpha) = \frac{2 \tan(\alpha)}{1 - \tan^2(\alpha)}$$

Ángulo mitad

$$\sin\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1 - \cos(\alpha)}{2}}$$

$$\cos\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1 + \cos(\alpha)}{2}}$$

$$\tan\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1 - \cos(\alpha)}{1 + \cos(\alpha)}}$$

Sumas y Restas de senos y cosenos

$$\sin(\alpha) + \sin(\beta) = 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \quad \sin(\alpha) - \sin(\beta) = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

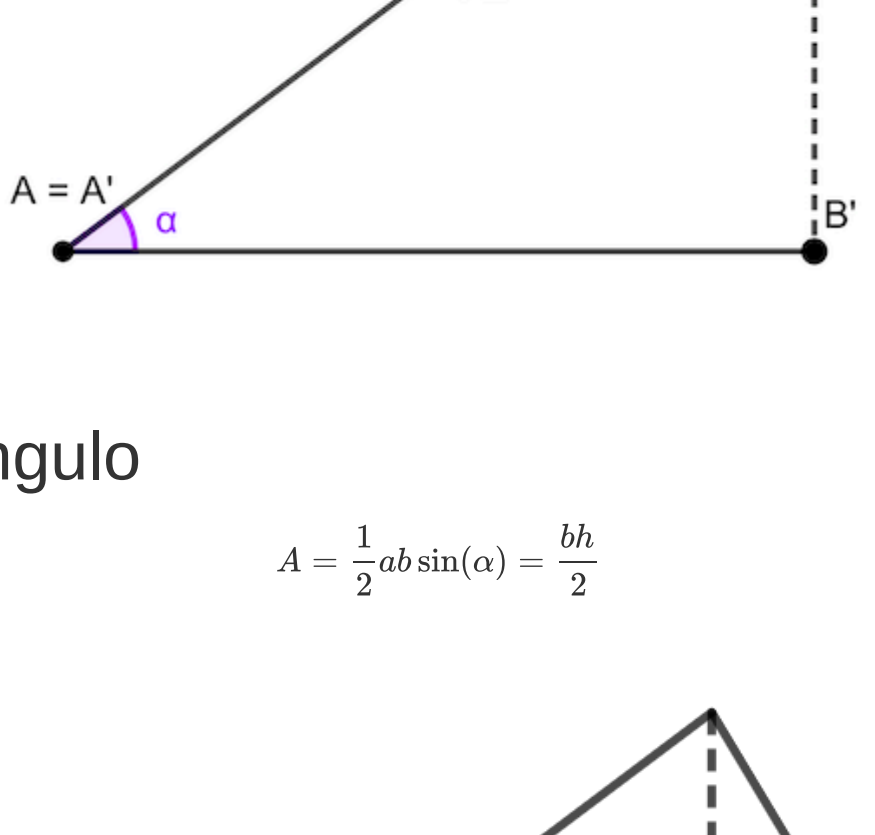
$$\cos(\alpha) + \cos(\beta) = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \quad \cos(\alpha) - \cos(\beta) = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

Valores del seno, coseno y tangente usuales

Radianes	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
Grados	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	∞

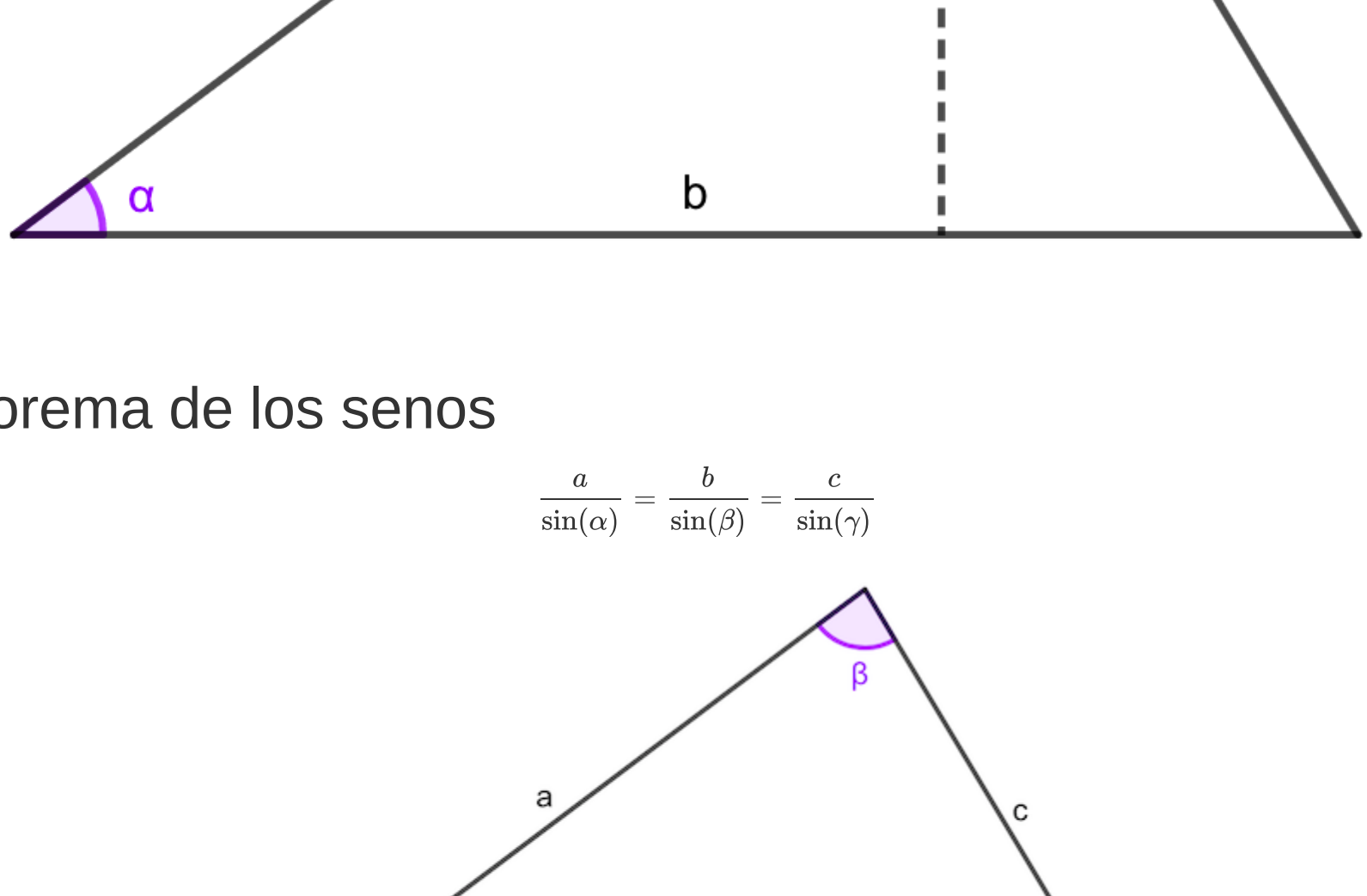
Proyección del segmento $\overline{AB}$ sobre una recta r

$$A'B' = AB \cos(\alpha)$$



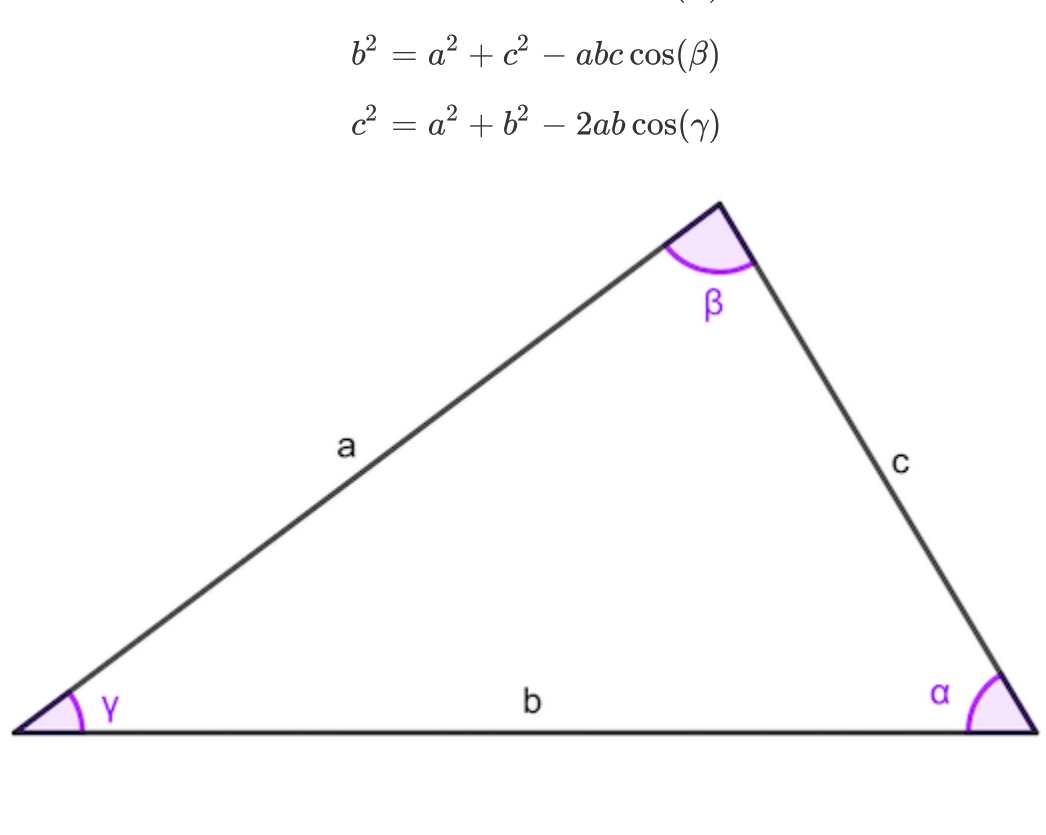
Área de un triángulo

$$A = \frac{1}{2} ab \sin(\alpha) = \frac{bh}{2}$$



Teorema de los senos

$$\frac{a}{\sin(\alpha)} = \frac{b}{\sin(\beta)} = \frac{c}{\sin(\gamma)}$$

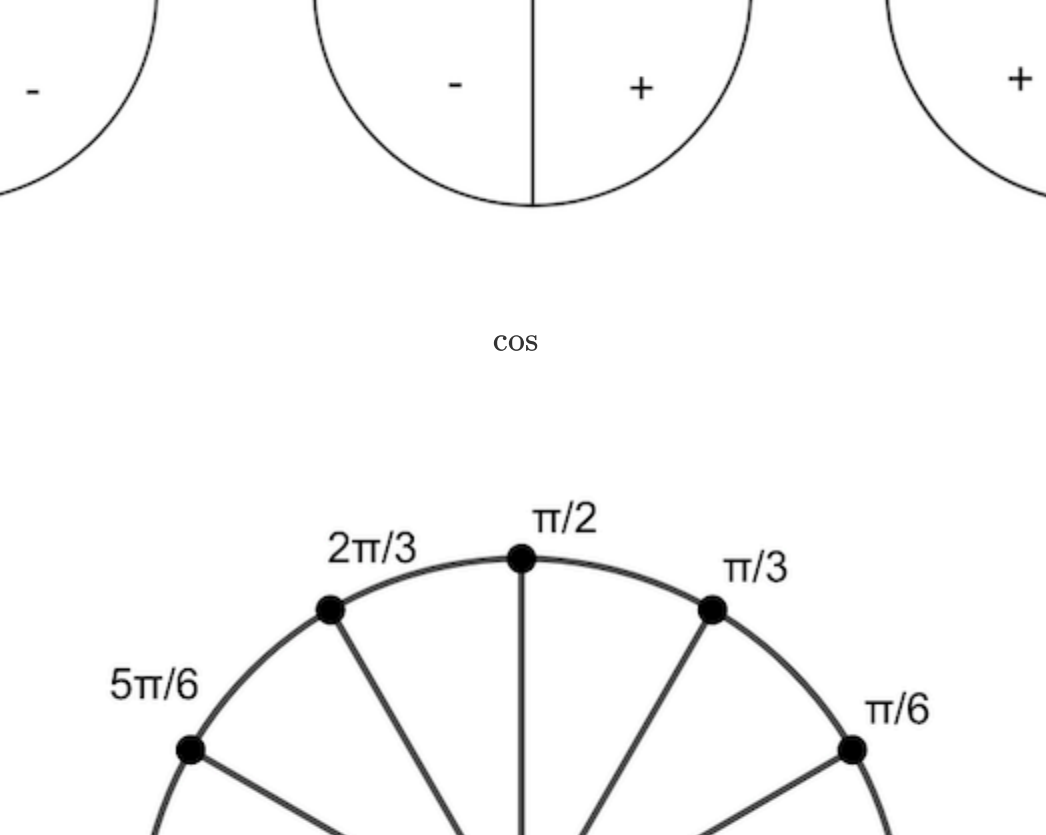


Teorema del coseno

$$a^2 = b^2 + c^2 - 2bc \cos(\alpha)$$

$$b^2 = a^2 + c^2 - 2ac \cos(\beta)$$

$$c^2 = a^2 + b^2 - 2ab \cos(\gamma)$$



Más sobre seno, coseno y tangente

